

Do 20 km of tunnels manage flooding?

Testing a public claim about Malta's flood-relief network against official sources and rainfall data

THE CLAIM UNDER REVIEW

“...they avoid flooding and damage caused by waters every time it rains.”

Attributed by TVM News to Mario Ellul, Public Works Department, 4 July 2025.
Said alongside a plan to extend the network from 16 km to a 20 km total.

VERDICT: NOT SUBSTANTIATED

Plausible by design; effect not shown

Version 1.1 · 6 October 2026

Status: right of reply on hold until the comparison is read

Prepared from public sources and open data.

Repository: github.com/leandergrech/Mizien

MIZIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

On 4 July 2025 Malta’s Public Works Department said its flood-relief canals, tunnels and culverts are being extended “from 16 km to 20 km”, and its director of the Coastal and Stormwater Unit, Mario Ellul, said these systems “**avoid flooding and damage caused by waters every time it rains.**” (TVM News attributes this to him). A MaltaToday headline, which we use only to locate the claim, reads “20 kilometres of underground tunnel helped Malta manage its flooding problems”; that is headline wording, not the department’s, and is not rated. We tested the department’s own statements against official sources and daily rainfall data.

- 1

The department’s own figures separate what exists from what is planned. It describes 16 km as in use and a further 4 km as still to be built, to reach a total of 20 km [1]. A 2020 news report also gave about 16 km [4]. We do not rate the headline’s “20 km” (Figure 1).
- 2

Design capacity is documented; outcomes are not. The European Commission describes a system that covers 65 km², is designed for a storm with a 5-year return period and is expected to benefit about 165,000 people [3]. In the sources we searched (Method) we found no published multi-year count of flood incidents, insurance claims or road closures before and after the works; the Energy and Water Agency says it compared the effect by modelling, but we found no results in its published plan [9].
- 3

Flooding has continued, and new tunnels are being planned. A news report of 15 September 2020 describes flooded roads in the project’s own localities [4]. In 2022 the department planned two more tunnels, at Mosta and Mater Dei, for flooding that persists in those places [2].
- 4

“Every time it rains” is stronger than the evidence. A system built for a 5-year storm will be exceeded by larger storms, and no source shows how often it prevents flooding. Our one rain gauge cannot test this (Figure 2).
- 5

Verdict: not substantiated (moderate confidence). The tunnels exist and are designed to reduce flooding, so an effect is plausible. The verdict rests on two sub-claims: the effect in practice is not shown (C), and “every time it rains” is stronger than the evidence (D).



WHAT WOULD MOVE THE VERDICT UP A published series of flood incidents (emergency calls, road closures or insurance claims) for the tunnelled catchments, showing a fall after the works, or the Energy and Water Agency’s before-and-after comparison [9] showing that flooding is avoided in storms up to the design level. Either would move the verdict to <i>Largely supported</i> .	WHAT WOULD MOVE IT DOWN Evidence that the same localities flooded as often or as badly after the works in storms below the design level. That would move the verdict to <i>Contradicted</i> .
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1 The claim and what we could verify

The check began from a MaltaToday headline, “20 kilometres of underground tunnel helped Malta manage its flooding problems” [5]. That page, which carries no speaker we could read, is refused to scripts (HTTP 403), so we saw only its headline and a search summary. A headline is not the speaker’s wording and we do not rate it. We therefore looked for the public statement behind it. The nearest in date and wording is a press release of the Parliamentary Secretariat for Public Works of 4 July 2025, reported with direct quotes by the public broadcaster’s news service, TVM News [1]. The release itself is on gov.mt, which refuses scripts [6]. TVM publishes its English text as a translation from Maltese, so wording below is TVM’s English rendering.

What was said	Form	Our access
The department “is working to increase from 16 km to 20 km the infrastructure of canals, tunnels and culverts in use to reduce flooding”	TVM’s summary, no quotation marks [1]; the next paragraph, also indirect speech, says Ellul explained that extra works would bring “the total to 20 km”	Read in full
“These are very important systems for the country, particularly for the Birkirkara localities, because they avoid flooding and damage caused by waters every time it rains.”	Pull-quote attributed by TVM to Mario Ellul, Public Works Department; the page prints no quotation marks [1]	Read in full
“As a department, we are making sure the underground infrastructure, including tunnels and culverts, is cleaned and maintained regularly”	Pull-quote attributed by TVM to Parliamentary Secretary Omar Farrugia; no quotation marks printed [1]	Read in full
“20 kilometres of underground tunnel helped Malta manage its flooding problems”	Headline of a MaltaToday article, undated [5]	Headline and search summary only

FAIRNESS NOTE

The department’s wording is about the network’s purpose and maintenance: it says the systems are meant “to reduce flooding” and that maintenance supports them. The stronger form (20 km of tunnel already helping) comes from a headline whose text we could not read in full. We test the department’s wording and rate only that.

2 Method

Question. Are there about 20 km of flood tunnels, and is there evidence that they have helped Malta manage flooding?

Evidence. We read the department’s statement as reported by TVM News [1, 2], the European Commission’s project page [3] and a 2020 news report [4]. For rainfall we used the daily record of the Luqa gauge from NOAA’s Global Historical Climatology Network [7], 1990–2025. We computed annual maxima for years with at least 330 daily readings and an empirical 5-year level (Weibull plotting position), with a script (*tools/cc-045-report/calc.py*). We searched Crossref and OpenAlex for a peer-reviewed evaluation of the Maltese tunnels and found none; OpenAlex was rate-limited on the day.

Search for flood-incident data (6 October 2026). The verdict depends on whether a before-and-after record of flooding exists, so we searched for one. Web searches, run for the Civil Protection Department, insurers, parliamentary questions, the National Statistics Office and the Energy and Water Agency (EWA), returned no incident series; they led us to four documents we opened: EWA’s Preliminary Flood Risk Assessment (2019) [8], its 2nd Flood Risk Management Plan (2024) [9], its flood hazard and risk maps report (2020) [10] and the Insurance Association Malta’s page on Storm Harry (2026) [11]. We searched the three EWA texts by keyword and read the sections on the tunnel project and on past floods; we did not read them end to end. They hold modelled hazard maps and one-event figures, not a multi-year count of incidents. We did not browse the Civil Protection Department, National Statistics Office, parliamentary-questions or news archives directly, and the Public Works Department page on rainwater relief returned HTTP 403. We therefore say only that we *found* no such series, not that none exists. The search log is in *literature/CC-045/notes.md*.

Grades. The department statement is grade D (assertion, no data) for effect and C for length. The Commission page is grade C (official description, design figures). The rainfall record is grade C (official observations), used as context only.

3 What the sources say

Length. The Commission’s page does not give a total length; it names four basins and nine localities [3]. Lovin Malta, in 2020, wrote of “some 16 kilometres of mostly underground tunnels” and €54 million [4]. The department in 2025 gives 16 km in use and 20 km as the planned total [1]. The department’s own count includes canals and culverts, not tunnels only, so “20 km of underground tunnel” is also a looser description than the department’s.

Design and expected benefit. The Commission says the infrastructure “covers 65 km² of the Maltese territory and is capable of handling a storm with a return period of 5 years”, and that “experts predict” it benefits some 165,000 residents [3]. These are design and forecast figures from 2013, before the works finished.

What continued afterwards. On 15 September 2020 Lovin Malta reported flooded roads, power cuts and overflowing drains and quoted the minister blaming an unusually heavy downpour and unfinished road works [4]. In 2022 the department said it would dig two further tunnels, one near Mater Dei Hospital and one under Valletta Road in Mosta, to deal with flooding in those places [2]. Both are consistent with a network that reduces flooding where it reaches and leaves some places exposed; neither measures how much it reduced.

What the Energy and Water Agency says. The Agency’s 2019 assessment says the project was designed for 5-year rainfall events, “thereby significantly reducing” their consequences in four sub-catchments, and that it also increased resilience to rarer events [8]. That is an official statement without incident data. Its 2nd Flood Risk Management Plan (2024) records that a comparison “ante and post-operam” of flood hazard and risk was carried out and shared with stakeholders [9]. The plan text we searched gives no results from that comparison, and the hazard maps are predictive: they do not use past floods except to validate them [10]. The comparison is a lead we have not seen.

4

What the data show

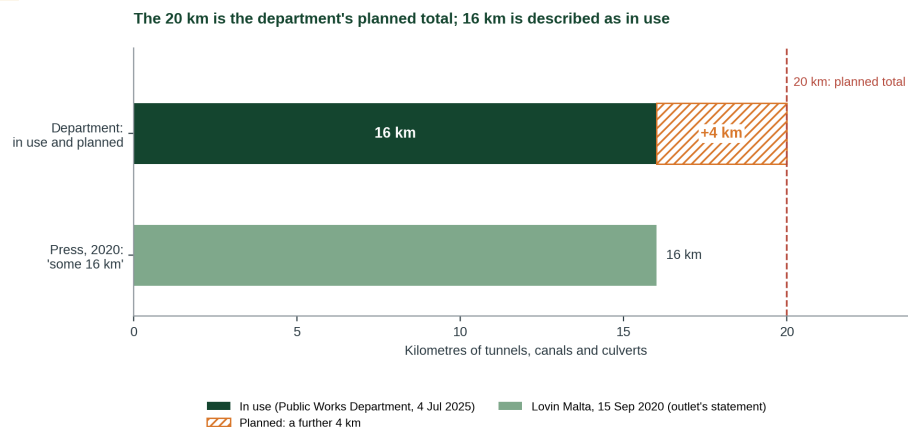
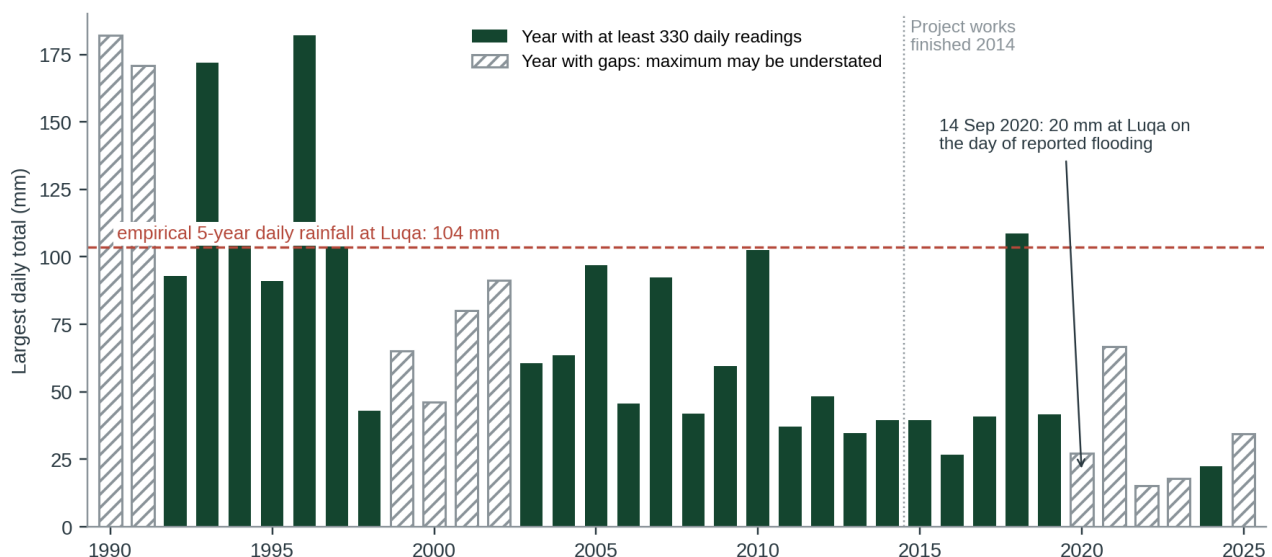


Figure 1. Length of the flood-relief network as stated by the department (2025) and in a 2020 news report [1, 4]. Values in data/cc-045/lengths.csv.

One gauge, 36 years: it cannot show whether the tunnels reduced flooding



Source: NOAA NCEI GHCN-Daily, Luqa (MT000016597), daily precipitation, retrieved 6 Oct 2026. Return level: empirical (Weibull), 25 complete years. A single gauge about 6 km from the Birkirkara tunnel; the project's design storm is defined for each catchment, so the line is context, not a test.

Figure 2. Largest daily rainfall each year at Luqa, 1990–2025, and the empirical 5-year level (104 mm) [7]. Hatched years have gaps. All values in data/cc-045/luqa_annual.csv and checks.csv.

Reading across the data

- **The length differs by source.** 16 km in use and 20 km planned (department, 2025) [1]; about 16 km (press, 2020) [4]. The claim's 20 km matches only the planned total.
- **The rainfall record cannot test the effect.** The gauge is about 6 km from the Birkirkara tunnel. Luqa recorded 20.1 mm on 14 September 2020, the day before Lovin Malta's report of flooding [4], below the 2-year level of 59 mm: localised downpours are not captured by one gauge.
- **The record is incomplete after 2019.** 2020–2023 and 2025 have many missing days, so only six years from 2015 are complete. We therefore make no before-and-after comparison of storms.

- **One storm at Luqa since 2014 exceeded the empirical 5-year level:** 108.7 mm on 10 February 2018. We found no sourced account of flooding that day, so we draw no conclusion from it.

5

Where the evidence points different ways

Do the tunnels reduce flooding?		NOT SHOWN
EVIDENCE FOR THE CLAIM'S IMPRESSION Design: a 5-year storm capacity over 65 km² and an expected 165,000 beneficiaries [3]. The department says the systems “avoid flooding” and spends a team on maintenance [1]. Reservoirs store rainwater for farmers [1].	EVIDENCE AGAINST Flooded roads in the project’s localities in September 2020 [4]; two further tunnels planned in 2022 for places that still flood [2]. No before-and-after incident data have been published in the sources we searched (Method).	
WHY THEY DIFFER, AND WHAT IT MEANS HERE A system can reduce flooding in many storms and still be overwhelmed by a storm above its design level, or by water arriving where no culvert reaches. Reports of flooding therefore do not show failure, and the department’s “every time it rains” cannot be tested without incident data. That is the gap behind the verdict.		

How long is the network?		NOT RATED
EVIDENCE FOR THE CLAIM'S IMPRESSION The department: 16 km in use, 20 km as the planned total [1]. A MaltaToday headline says 20 km [5].	EVIDENCE AGAINST 16 km in use (department) [1]; about 16 km (Lovin Malta, 2020) [4].	
WHY THEY DIFFER, AND WHAT IT MEANS HERE The figures agree once planned and operating lengths are separated, as the department itself separates them. The headline’s “20 km ... helped” merges them; that is headline wording, which we do not rate.		

6

Testing the claim

Sub-claim	What the evidence shows	Rating
A. 16 km in use, 20 km as the planned total (department's own figures)	Stated as 16 km in use, 4 km planned, 20 km total (department, 4 Jul 2025) [1]; about 16 km in 2020 [4]. The MaltaToday headline's “20 km ... helped” is headline wording, not rated.	NOT RATED
B. They are designed to reduce flooding	65 km², 5-year storm, 165,000 expected beneficiaries (Commission, 2013) [3].	SUPPORTED
C. The infrastructure is in use “to reduce flooding” (department, as summarised by TVM)	Plausible from design; no before-and-after incident data found in the sources searched. Flooding reported in project localities in 2020 [4] and new tunnels planned in 2022 [2]. The Agency's own comparison is not published in what we read [9].	NOT SHOWN
D. The systems “avoid flooding ... every time it rains” (attributed to Ellul, “particularly for the Birkirkara localities”)	A 5-year design is exceeded by larger storms; no source shows how often flooding is prevented [1, 3, 4, 8].	OVERSTATED

7 Verdict and requests for evidence

VERDICT

NOT SUBSTANTIATED

Plausible from design, but no outcome data show the effect and “every time it rains” exceeds a 5-year design.
Confidence: moderate.

Why. The verdict rests on sub-claims C and D. (1) The network exists and was built to reduce flooding (B). (2) We found no published flood-incident series before and after the works in the sources we searched, so the effect in practice (C) rests on design figures and the department’s statement. (3) The strongest wording, “every time it rains” (D), is above the 5-year design level. (4) The length (A) is not part of the verdict: the department’s figures separate 16 km in use from the 20 km planned total, and the headline is not rated.

What this verdict does not say. It does not say the tunnels do not work, that flooding is worse, or that anyone misdescribed the project. The published record cannot show their effect either way.

Evidence we are asking for

- ☐ Flood incidents (emergency calls, road closures, insurance or compensation claims) for Birkirkara, Msida, Gżira, Qormi, Marsa, Żabbar and Marsaskala, 2005 to date.
- ☐ The Storm Water Master Plan and the design-storm basis of each catchment, and the operating length by tunnel.
- ☐ Hourly rainfall for flood days from the Malta Meteorological Office and the department’s own gauges.
- ☐ The gov.mt release of 4 July 2025, and the Energy and Water Agency’s ante and post-operam comparison of the flood relief project [9].

RIGHT OF REPLY

On hold. The Energy and Water Agency’s before-and-after comparison of the project (2nd Flood Risk Management Plan, measure FLD1) has not been read and could change the verdict. Once it is read, this draft will be sent to the Public Works Department with a fixed deadline for reply; until then it should not be circulated elsewhere.

8 Limitations

- We could not read the MaltaToday article or the gov.mt release (HTTP 403); we rely on TVM News’s account of the release and on the article’s headline and a search summary (♦ second-hand).
- TVM’s English text is a translation of a statement in Maltese. TVM prints the Ellul and Farrugia sentences as attributed pull-quotes without quotation marks, so we quote TVM’s text, not a transcript.
- The rainfall record is one gauge, with gaps after 2019; it gives context, not a test. The 5-year level is empirical (25 complete years) and is not the project’s design storm.
- We did not find peer-reviewed work on the Maltese tunnels. We found no flood-incident data in a partial search (Method), so the report cannot say whether flooding fell. The Agency’s own before-and-after comparison [9] was not seen.

References

- [1] TVM News (4 Jul 2025). 2,400 tonnes of waste removed from 16 km of stormwater tunnels. Reports the Parliamentary Secretariat for Public Works release; statements of Mario Ellul and Omar Farrugia.
<https://tvmnews.mt/en/news/2400-tonnes-of-waste-removed-from-16-km-of-stormwater-tunnels/>
- [2] TVM News (28 Aug 2022). Underground tunnels being excavated to alleviate flooding.
<https://tvmnews.mt/en/news/underground-tunnels-being-excavated-to-alleviate-flooding/>
- [3] European Commission, Inforegio (15 Oct 2013). Fighting floods in Malta (National Flood Relief Project).
https://ec.europa.eu/regional_policy/projects/projects-database/fighting-floods-in-malta_en
- [4] Lovin Malta (15 Sep 2020). An underground tunnel and millions in road investment later, Malta is still flooding.
<https://lovinmalta.com/opinion/analysis/an-underground-tunnel-and-millions-in-road-investment-later-but-malta-is-still-flooding/>
- [5] MaltaToday (undated). 20 kilometres of underground tunnel helped Malta manage its flooding problems. ♦ Headline and search summary only (HTTP 403). https://www.maltatoday.com.mt/environment/nature/107804/20_kilometres_of_underground_tunnel_helped_malta_manage_its_flooding_problems
- [6] Government of Malta, Department of Information (4 Jul 2025). Press release by the Parliamentary Secretariat for Public Works: over 2,400 tonnes of waste collected annually from 16 kilometres of stormwater tunnels. ♦ Not read (HTTP 403); cited through [1].
<https://www.gov.mt/en/Government/DOI/Press%20Releases/Pages/2025/07/04/pr251213en.aspx>
- [7] NOAA National Centers for Environmental Information. Global Historical Climatology Network daily, station MT000016597 (Luqa), precipitation. Retrieved 6 Oct 2026. <https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=MT000016597&dataTypes=PRCP&format=csv>
- [8] Energy and Water Agency (2019). Preliminary Flood Risk Assessment for the Malta River Basin District 2019. Keyword search and reading of the National Flood Relief Project sections; not read end to end. <https://www.energywateragency.gov.mt/wp-content/uploads/2020/09/Preliminary-Flood-Risk-Assessment-for-the-Malta-River-Basin-District-2019.pdf>
- [9] Energy and Water Agency (2024). 2nd Flood Risk Management Plan for the Malta River Basin District, measure FLD1. Keyword search; not read end to end.
<https://energywateragency.gov.mt/wp-content/uploads/2024/02/2nd-Flood-Risk-Management-Plan-Final.pdf>
- [10] Energy and Water Agency (2020). Flood Hazard Maps and Flood Risk Maps for the Malta River Basin District (final draft). Keyword search and reading of the methodology section.
https://energywateragency.gov.mt/wp-content/uploads/2022/09/FHM-and-FRM_11_12_2020_Final_Draft.pdf
- [11] Insurance Association Malta (24 Mar 2026). Storm Harry leaves EUR 12m in damage claims, majority from flooding. Single-event figures; the full Sunday Times of Malta article was not read.
<https://maltainsurance.org/general/storm-harry-leaves-e12m-in-damage-claims-majority-from-flooding/>

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	6 Oct 2026	First issue. Draft pending right of reply from the Public Works Department.
1.1	6 Oct 2026	Corrections after the 6 Oct 2026 audit: the claim and sub-claims now restate the department’s own wording (the MaltaToday headline and sub-claim A are not rated); the verdict is stated to rest on C and D; the unsourced word “solved” removed; the Ellul and Farrugia sentences described as TVM pull-quotes without quotation marks; a logged search for flood-incident data and the Energy and Water Agency’s documents added; layout fixed. Verdict unchanged. Right of reply on hold until the Energy and Water Agency’s before-and-after comparison is read.